Biosignal Metrics Guide

EEG • PPG / HRV • IMU • Temperature

Part 1: Plain-English guide — what each metric means and how it shows up in real life.

Part 2: The big picture — how metrics combine into real biological outputs, with reasoning.

Part 3: Full technical reference — formulas, worked examples, and clinical significance.

Part 1 — Plain-English Guide

No formulas here. Just what each metric actually tells you, and where you've probably already seen it at work — in a smartwatch, a meditation app, a hospital, or a doctor's office.

1. Power Spectral Density (PSD)

EEG — the foundation

What it is: Think of a raw brainwave signal as a messy chord being played on a piano all at once. PSD is the process of figuring out exactly which individual notes (frequencies) are being played, and how loudly each one is being played.

Where you'll meet it in real life: *You never see this number directly, but it's the engine every other brain metric below runs on. It's the reason a $200 EEG headband gives you a clean, stable reading instead of random static.*

2. Band Power (Delta, Theta, Alpha, Beta, Gamma)

EEG — "how much of each brainwave type"

What it is: Once you know which notes are playing, band power just adds up how loud the notes are in five named ranges — slow, sleepy ones (delta/theta) up through fast, alert ones (beta/gamma).

Where you'll meet it in real life: *This is the actual math behind the "focus" and "calm" scores on Muse and Emotiv headbands, the deep-sleep percentage on sleep trackers, and the standard test doctors use to flag drowsiness or cognitive impairment in an awake patient.*

3. Spectral Centroid

EEG — one number for "how alert is this brain right now"

What it is: Instead of tracking five separate bands, this squashes the whole spectrum into a single number: the "center of gravity" of the brain's activity. A higher number means activity is skewed toward fast/alert waves; a lower number means it's skewed toward slow/drowsy waves.

Where you'll meet it in real life: *This is the kind of single trend-line number used in driver-drowsiness alert systems in cars — simpler to watch in real time than five separate charts.*

4. Frontal Alpha Asymmetry (FAA)

EEG — a mood-lean indicator

What it is: Compares the calm-brainwave activity on the left side of your forehead to the right side. More activity dominance on the left is linked to "approach" mood (motivated, positive); more on the right is linked to "withdrawal" mood (avoidant, negative).

Where you'll meet it in real life: *This is a real research marker in depression studies — reduced left-frontal activity shows up repeatedly in people with depressive symptoms. It's one of the more scientifically credible single-pair EEG measurements out there.*

5. Theta/Beta Ratio (TBR)

EEG — a popular but debated "attention" score

What it is: A simple ratio of slow brainwaves to fast brainwaves. Historically, a higher ratio was thought to mean "less focused / more inattentive."

Where you'll meet it in real life: *This is marketed heavily in consumer neurofeedback and attention-training products as an ADHD-related score. Worth knowing: newer research has cast real doubt on how reliable this ratio actually is, so treat any product claim built on it with some skepticism.*

6. Alpha Suppression (Eyes Open vs. Closed)

EEG — the equipment sanity check

What it is: A healthy brain shows a strong calm rhythm (alpha) with eyes closed, which should drop noticeably the moment the eyes open. This metric just measures how much it drops.

Where you'll meet it in real life: *Every EEG lab and every consumer headband relies on this as a built-in quality check. If this pattern doesn't show up, the problem is almost always a bad electrode connection — not the person wearing it.*

7. Heart Rate (from PPG)

PPG — your smartwatch's heart rate number

What it is: PPG is the little light sensor on the back of a smartwatch or fitness band that shines light into your skin and reads the pulses in your blood flow. This metric turns that raw pulsing signal into a simple beats-per-minute number.

Where you'll meet it in real life: *This is, quite literally, the heart rate number displayed on every Apple Watch, Fitbit, Whoop, Oura ring, and cheap drugstore fitness tracker.*

8. HRV: LF, HF, and LF/HF Ratio

PPG / Heart-rate variability — the "stress" and "recovery" score

What it is: Your heartbeat doesn't tick like a metronome — the tiny gaps between beats speed up and slow down constantly. This metric studies the pattern of that variability to estimate how much your nervous system is in "rest and digest" mode versus "fight or flight" mode.

Where you'll meet it in real life: *This is the actual math behind the stress score, recovery score, and readiness score on Whoop, Oura, and Fitbit. Hospitals also use a version of it clinically — low HRV after a heart attack is a known warning sign for worse outcomes.*

9. Postural Sway Power

IMU (motion sensor) — balance and fall-risk

What it is: Using a motion sensor (like the one in a phone or a wearable), this measures the tiny, often invisible wobbles a person makes while just standing still.

Where you'll meet it in real life: *This exact measurement is used in real fall-risk assessments for elderly patients and in monitoring balance problems in Parkinson's disease — more wobble at low frequencies is linked to a higher chance of falling.*

10. Skin Temperature Rate of Change

Temperature — a supporting stress clue

What it is: Just tracks how quickly your skin temperature is rising or falling. When your body goes into a stress response, blood vessels near the skin constrict and your hands/feet get measurably colder, fast.

Where you'll meet it in real life: *This is why "cold hands" is a real physiological stress sign, not just an expression. It's used as a backup signal in stress-tracking wearables and in polygraph-style tests — but never trusted alone, since a cold room fools it just as easily as real stress does.*

11. Composite Stress Index (CSI)

Everything combined — the single score you actually see

What it is: This takes several of the metrics above (attention ratio, heart-rate-variability ratio, balance wobble, relaxation signal) and blends them into one overall number using a weighted formula.

Where you'll meet it in real life: *This is essentially how a smartwatch or wearable turns a pile of messy raw sensor data into the one simple "Stress: 62/100" number you glance at on your screen.*

Part 2 — The Big Picture: How Raw Signals Become a Real Biological Verdict

Part 1 explained each individual metric on its own. But no single metric is trusted alone in this system. The real value comes from **combining several metrics into one biological conclusion** — the way a doctor doesn't diagnose you from one symptom, but from a pattern of symptoms that all point the same direction. This section shows, for each real-world biological output the system produces, exactly which raw metrics feed into it, the step-by-step derivation, and — most importantly — **why** that combination is biologically justified rather than arbitrary.

OUTPUT 1: Vigilance / Drowsiness State

What this output tells you: Is the brain currently in an alert, engaged state, or drifting toward drowsy / low-vigilance?

How it's derived (step by step):

1. Take the raw EEG signal and run Power Spectral Density (PSD) to see which frequencies are present.
2. Calculate Band Power for the slow bands (delta, theta) and fast bands (beta, gamma).
3. Calculate the Spectral Centroid — one number for whether the overall spectrum leans slow or fast.
4. Cross-check the reading using the Alpha Suppression test (eyes open vs. closed) to confirm the electrodes are actually working and the reading is trustworthy.
5. Combine slow/fast band ratio + spectral centroid into a single vigilance score.

Why this combination makes biological sense: A drowsy or low-vigilance brain physically shifts its dominant electrical rhythm toward slower frequencies (theta/delta) and away from fast, active rhythms (beta/gamma) — this is a well-established, decades-old finding in sleep and attention research. Using two different measures of "where the energy sits" (band power AND spectral centroid) instead of just one protects the output from being thrown off by noise in any single calculation, and the alpha-suppression check guards against reporting a confident-sounding number from a badly-fitted sensor.

What the final readout looks like: *"Vigilance: 74/100 — alert, fast-wave-dominant pattern; sensor check passed."*

Parameter-to-Output Mapping — Vigilance / Drowsiness State

Parameter	Typical / Range (units)	Proportionality → Output	Physiological Effect in Humans
Slow/Fast Power Ratio ((delta+theta) / (beta+gamma))	Alert: 0.4–1.2 Drowsy: 1.5–3.0 Sleep-onset: >3.5	**Inverse** — ratio ↑ ⇒ Vigilance score ↓	Thalamocortical rhythms shift to slower frequencies as wakeful arousal drops — the classic EEG signature of drowsiness and sleep onset.
Spectral Centroid	Alert: 12–20 Hz Relaxed: 8–12 Hz Drowsy: 5–9 Hz	**Direct** — centroid ↑ ⇒ Vigilance score ↑	Captures the overall balance of fast vs. slow cortical activity in one number; higher values reflect more active, engaged cortical processing.

Parameter	Typical / Range (units)	Proportionality → Output	Physiological Effect in Humans
Alpha Suppression % (validity gate)	Healthy: 40–70% suppression Poor/invalid: <20%	**Not proportional** — acts as a confidence gate, not a scoring input	Confirms a genuine occipital alpha rhythm is present and electrodes are contacting properly. Below threshold, the other two readings are flagged low-confidence rather than mathematically adjusted.

Differentiators — what unique information each parameter adds:

- **Slow/Fast Ratio** differentiates coarse frequency-band dominance — the "which half of the spectrum wins" signal.
- **Spectral Centroid** differentiates fine-grained spectral tilt — catches partial or gradual shifts the ratio alone can miss (e.g. a slight broadening of beta without fully crossing into a new band).
- **Alpha Suppression** differentiates signal *validity* from signal *value* — it doesn't say how alert someone is, it says whether the other two numbers can be trusted at all.

OUTPUT 2: Attention / Focus Tendency

What this output tells you: A trend indicator of sustained attention versus mind-wandering, expressed as a single trackable score.

How it's derived (step by step):

1. From the same PSD, isolate theta band power and beta band power specifically.
2. Compute the Theta/Beta Ratio (TBR).
3. Track this ratio over time rather than reading a single snapshot in isolation.

Why this combination makes biological sense: The theory behind TBR is that theta (slow, internally-focused activity) rising relative to beta (fast, externally-focused activity) reflects a shift away from task engagement. This is why it's used the way it is here: **as a trend to watch, not a diagnosis to declare** — recent research has genuinely weakened confidence in TBR as a standalone attention marker, so this output is reported with that caveat attached rather than presented as a hard fact.

What the final readout looks like: *"Attention trend: rising theta/beta ratio over the last 10 minutes — possible attention dip (low-to-moderate confidence)."*

Parameter-to-Output Mapping — Attention / Focus Tendency

Parameter	Typical / Range (units)	Proportionality → Output	Physiological Effect in Humans
Theta Power (4–8 Hz)	Typical resting: 8–20 μV²	**Direct** — Theta ↑ (numerator) ⇒ TBR ↑ ⇒ Attention score ↓	Reflects internally-directed cognition, mind-wandering, and drowsiness; rises during daydreaming and reduced task engagement.
Beta Power (13–30 Hz)	Typical resting: 5–15 μV²	**Inverse** — Beta ↑ (denominator) ⇒ TBR ↓ ⇒ Attention score ↑	Reflects active concentration and sensorimotor engagement — but also rises with anxiety and muscle tension, a known confound worth flagging.

Parameter	Typical / Range (units)	Proportionality → Output	Physiological Effect in Humans
Theta/Beta Ratio (TBR)	Population avg. (eyes-open rest): 1.5–2.0 Inattention-associated: >2.5–3.0 Engaged: <1.2	**Direct** — TBR ↑ ⇒ reported inattention risk ↑ (disputed)	Historically proposed as reflecting cortical under-arousal during inattentive states; reliability as a standalone marker is contested in recent replication studies.

Differentiators — what unique information each parameter adds:

- **Theta power** differentiates the "drifting inward" component of attention on its own.
- **Beta power** differentiates the "actively engaged" component — but must be read alongside context, since beta rises for both genuine focus and anxious arousal.
- **TBR** differentiates a trend ratio from either raw band alone — it smooths out cases where both bands rise or fall together for unrelated reasons (e.g. a general change in electrode contact quality).

OUTPUT 3: Mood Lean (Approach vs. Withdrawal)

What this output tells you: Whether the brain's current activity pattern leans toward an approach-motivated (positive/engaged) state or a withdrawal-motivated (negative/avoidant) state.

How it's derived (step by step):

1. Take alpha band power separately from the left-frontal electrode (F3) and the right-frontal electrode (F4).
2. Log-transform each value (this corrects for the fact that raw power values are skewed and not directly comparable on a straight scale).
3. Subtract: FAA = ln(alpha at F4) – ln(alpha at F3).

Why this combination makes biological sense: This isn't an arbitrary left-vs-right comparison — it's built on a specific, well-replicated neuroscience finding (Davidson's frontal asymmetry model) that the left and right frontal cortex play different roles in emotional motivation. Because it only ever compares a person to *their own* left-vs-right baseline, it sidesteps a lot of the person-to-person variability that makes single absolute EEG numbers hard to trust.

What the final readout looks like: *"FAA = +0.47 — mild left-frontal dominance, consistent with an approach-motivated state."*

OUTPUT 4: Autonomic Balance (Sympathetic "Fight-or-Flight" vs. Parasympathetic "Rest-and-Digest")

What this output tells you: Whether the nervous system, right now, is leaning toward an activated stress response or a calm, recovering state.

How it's derived (step by step):

1. From the heart-rate sensor (PPG), extract the RR-interval / IBI series (the exact timing between beats), then run that through its own PSD.
2. Calculate LF power and HF power from that heart-rate-variability spectrum.
3. Calculate LF/HF ratio.
4. Pull in Skin Temperature Rate of Change as a second, independent confirmation signal.

Why this combination makes biological sense: This is the clearest example of why the system never relies on one channel alone. A real stress response is a *whole-body* event: it changes heart-rate-variability patterns (captured by LF/HF) **and** constricts blood vessels near the skin, causing measurable cooling (captured by temperature rate of change) — at the same time, for the same underlying reason. When both signals move together in the expected direction, that agreement is far more trustworthy than either one alone; if they disagree, that's a flag that something else (like a cold room) is confounding the reading, not a real stress event.

What the final readout looks like: *"Autonomic state: sympathetic-leaning — LF/HF = 1.5, skin temperature falling 0.2°C/min — signals agree, moderate confidence stress event."*

Parameter-to-Output Mapping — Autonomic Balance

Parameter	Typical / Range (units)	Proportionality → Output	Physiological Effect in Humans
LF Power (0.04–0.15 Hz band)	Typical resting: 150–800 ms² (highly individual)	**Direct** — LF ↑ ⇒ LF/HF ↑ ⇒ Sympathetic-lean output ↑	Reflects a mix of sympathetic activity and baroreflex/blood-pressure regulation rhythms; rises with physical or mental stress and with standing posture.
HF Power (0.15–0.4 Hz band)	Typical resting: 150–800 ms²	**Inverse** — HF ↑ ⇒ LF/HF ↓ ⇒ Sympathetic-lean output ↓ (Parasympathetic ↑)	Reflects vagal (parasympathetic) tone tied to breathing rhythm; rises during calm states and slow, deep breathing.
LF/HF Ratio	Balanced: 1.0–2.0 Sympathetic-dominant: >2.0–3.0 Parasympathetic-dominant: <1.0	**Direct** — ratio ↑ ⇒ stress-lean output ↑	Summarizes the sympathetic/parasympathetic tug-of-war in one trend number; clinically debated as an absolute measure, more defensible as a within-person trend over time.
Skin Temperature Rate of Change	Stable/resting: ±0.05 °C/min Acute stress: –0.1 to –0.3 °C/min Recovery: +0.05 to +0.2 °C/min	**Inverse** — rate ↓ (more negative) ⇒ Sympathetic-lean output ↑	Peripheral vasoconstriction under sympathetic activation reduces blood flow to the skin, causing a measurable, if slower, temperature drop; confirms whether an HRV-based reading reflects a real whole-body event.

Differentiators — what unique information each parameter adds:

- **LF power** differentiates the "activation" side of the autonomic tug-of-war, though it's a blend of sources and not a pure sympathetic marker on its own.
- **HF power** differentiates the "calming" side, and is a cleaner, more specific vagal/parasympathetic marker than LF is for sympathetic activity.
- **LF/HF ratio** differentiates directionality — collapses two competing signals into one trend number, more useful for tracking change over time than for declaring an absolute state.
- **Skin temperature** differentiates time-scale: it's a slower (minutes), independent confirmation channel versus the faster, beat-to-beat HRV metrics — guarding against false positives caused by movement artifacts in the HRV signal alone.

*Reading note: every output above is a **trend indicator built from converging evidence**, not a medical diagnosis. Wherever the underlying science is still debated (TBR, LF/HF as a pure "stress" marker), that uncertainty is carried through to the final readout rather than hidden behind a confident-looking number.*

Part 3 — Technical Reference

Full formulas, what differentiates each metric's output, a worked numeric example, and real-world / clinical significance.

EEG

1. Power Spectral Density (Welch's method)

$$P_{xx}(f) = (1/K) \cdot \Sigma[k=1 \text{ to } K] |X_k(f)|^2$$

Differentiator: Each segment $X_k(f)$ is the FFT of a windowed, overlapping chunk of the signal; averaging K segments reduces the noisy/ripple artifact of a single raw periodogram.

Example: A 4-second EEG epoch at 256 Hz, split into 1-second windows (50% overlap) → 7 segments, each FFT'd, then averaged bin-by-bin.

Significance: This is the base calculation every other EEG metric below is built from.

2. Band Power

$$BP_{band} = \Sigma[f=f1 \text{ to } f2] P_{xx}(f) \cdot \Delta f$$

$$BP_{rel} = BP_{band} / \Sigma(\text{all bands}) BP \quad \text{(relative/normalized power)}$$

Differentiator: Which frequency range you sum over — delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), gamma (30–45 Hz).

Example: PSD bins 0.25 Hz apart; alpha band (8–13 Hz) bins sum to 45 μV²/Hz across 20 bins → alpha band power ≈ 45 × 0.25 = 11.25 μV².

Significance: Alpha power is the most clinically-used EEG marker — alpha suppression on eye-opening is a standard vigilance check; elevated theta/delta in awake adults flags drowsiness or cognitive impairment. Backbone of commercial EEG wearables (Muse, Emotiv) for "focus/calm" scoring.

3. Spectral Centroid

$$SC = \Sigma[i] f_i \cdot P(f_i) / \Sigma[i] P(f_i)$$

Differentiator: The "center of mass" of the spectrum — distinguishes a high-frequency-dominant signal (higher SC, more arousal/alertness) from a low-frequency-dominant one (drowsy/relaxed), even when band-power buckets look similar.

Example: Most power sits at 10 Hz with a tail to 20 Hz → SC lands around 12 Hz, a single number summarizing "where the energy sits" without separate band comparisons.

Significance: Used in fatigue-detection systems (driver drowsiness monitors) as a single scalar trend indicator instead of tracking 5 separate bands.

4. Frontal Alpha Asymmetry (FAA)

$$FAA = \ln(\alpha_{F4}) - \ln(\alpha_{F3})$$

Differentiator: Right (F4) minus left (F3), log-transformed alpha power. Positive FAA → left frontal activation dominance (approach motivation/positive affect); negative → right dominance (withdrawal motivation/negative affect).

Example: α_F4 = 8 μV^2, α_F3 = 5 μV^2 → FAA = ln(8) – ln(5) = 2.079 – 1.609 = **0.47** (left-hypoactive pattern, associated with depressive/withdrawal states in the literature).

Significance: A real diagnostic marker in depression research (reduced left frontal activity correlates with depressive symptomatology) — one of the more evidence-backed single-channel-pair EEG metrics.

5. Theta/Beta Ratio (TBR)

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TBR = BP_theta / BP_beta
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Example: Theta = 15 μV^2, Beta = 10 μV^2 → TBR = 1.5

Significance: Historically used as an ADHD biomarker (elevated TBR in inattentive states) — though its clinical reliability has been contested in later studies, it's still widely used in neurofeedback products as an attention proxy.

6. Alpha Suppression (eyes open vs. closed)

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ASM (%) = [(α_closed – α_open) / α_closed] × 100
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Example: —

Significance: The standard EEG cap-fitting/signal-quality sanity check used in every lab — if you don't see suppression on eye-opening, your electrodes are bad, not your subject.

PPG / HRV

7. Heart Rate from PPG (FFT-based)

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HR (bpm) = f_peak × 60
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Differentiator: f_peak = frequency (Hz) of the dominant peak in the PPG PSD, typically searched in the 0.8–3 Hz band.

Example: Peak at 1.2 Hz → HR = 72 bpm.

Significance: Direct measurement basis for wearable heart-rate displays.

8. HRV Frequency-Domain

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LF = ∫[0.04 to 0.15] P_RR(f) df

HF = ∫[0.15 to 0.4] P_RR(f) df

LF/HF = LF / HF
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Differentiator: Needs an RR-interval / IBI series, then FFT'd.

Example: LF = 450 ms^2, HF = 300 ms^2 → LF/HF = 1.5 (roughly balanced sympathovagal tone; ratios >2 lean sympathetic-dominant/stressed, <1 lean vagal-dominant/relaxed — though this ratio's clinical validity is also debated in newer HRV literature, worth flagging as such rather than stated as settled fact).

Significance: The physiological basis of every "stress score" on a Fitbit/Whoop/Oura — HF (vagal/parasympathetic) power dropping under stress is one of the more robust autonomic markers available. Also used clinically in cardiac risk stratification (reduced HRV post-MI predicts poorer outcomes).

IMU

9. Postural Sway Power

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SwayPower = ∫[0.1 to 1.0] P_accel(f) df
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Example: Summed PSD across 0.1–1 Hz on the mediolateral accel axis = 0.08 g²/Hz·Hz → feeds into a composite motor-stability input.

Significance: Used in fall-risk assessment for elderly patients (increased low-frequency sway power predicts fall likelihood) and in Parkinson's postural instability monitoring.

Temperature

10. Rate of Change (not spectral)

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dT/dt ≈ (T2 − T1) / (t2 − t1)
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Example: —

Significance: Peripheral (skin) temperature drops under acute sympathetic activation (vasoconstriction) — the mechanism behind why cold hands correlate with anxiety; used as a corroborating feature in polygraph-style and wearable stress systems, never a standalone marker because it's slow and confounded by ambient temperature.

Composite

11. Composite Stress Index (CSI)

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CSI = w1·TBR_norm + w2·(LF/HF)_norm + w3·SwayPower_norm − w4·HF_norm
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Differentiator: Weights w_i are tuned/validated against labeled datasets; worth stating the fusion structure generically in a paper rather than exposing exact weights if protecting an IP angle.

Example: —

Significance: Fuses multiple channels into a single actionable score.

Paper-writing note: a few of these metrics (TBR, and LF/HF as a pure "sympathetic marker") have real controversy in recent literature. Citing them as established rather than commonly-used-but-debated would weaken an academic paper's credibility — worth a sentence acknowledging that in a limitations section.